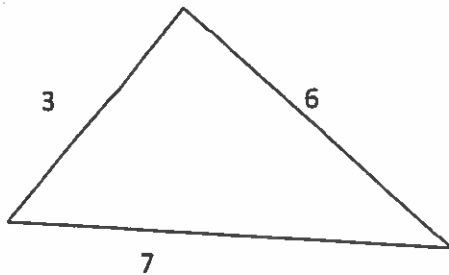


Tour de Maths : senior competition

Question 1

1000 dots are evenly spaced on the circumference of a circle. They are numbered from 1 to 1000 with dot 1000 opposite dot 500. Which dot is opposite 657?

Question 2



An ant walks around this triangle so that he is always 3 metres from the perimeter of the triangle. How far does he walk?

Question 3

Supply the next two numbers in the following sequence

1; 2; 4; -4; 7; 8; 10; -16; -; -

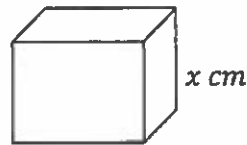
Question 4

Given : the following 6 numbers

11; 31; 19; 3; 10; 6

Three numbers are selected from the above set and added together. The remaining three numbers are also added together. These possible two sums are then multiplied to get a product. What is the maximum product?

Question 5



If a cube has a side $x \text{ cm}$. Find the value of this side if the volume of the cube is the same as the total surface area.

Question 6

Five straight lines are drawn. What is the maximum number of points of intersection?

Question 7

John can build a wall in 16 hours if he works alone. Jane can build the same wall in 12 hours if she works alone. If they work together they can build the wall in 8 hours, but because they sometimes get in each other's way, they build 16 bricks less per hour than they would if they did not get in each other's way. How many bricks are there in the wall?

Question 8

Consider

1
3 5 7
9 11 15 17
19 21 23 25 27 29 31

If the pattern continues

What is the value of the middle number in the 61st row?

Question 9

The average mark for 100 students in a maths test is 39%. The average mark for the students who passed the test is 60% and the average mark for the students who failed the test is 30%. How many students passed the test?

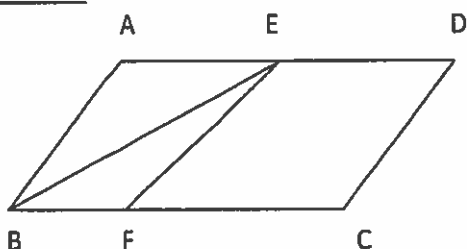
Question 10

If m is a solution to the equation $x^2 + 3x + 7 = 0$, find the numerical value of $2m^2 + 6m - 9$.

Question 11

If $\text{mod}(a;b)$ is the remainder when b is divided by a and $\langle x \rangle$ is the largest integer smaller than x

Find $\text{mod}(9; 17) - \langle -8.4 \rangle$

Question 12

ABCD is a parallelogram. E is the midpoint of AD and F is a point on BC such that $BF : FC = 1 : 3$

Calculate $\frac{\text{area of } \triangle BEF}{\text{area parallelogram } ABCD}$

Question 13

Determine m and n (both positive integers) such that

$$2^{20} + 2^{24} + 2^{25} + 2^n = m^2$$

Question 14

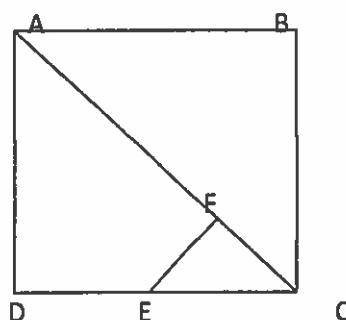
What is the smallest integer x , for which $\frac{15}{x-1}$ is an integer?

Question 15

The function $f(x)$ satisfies the equation

$$f(1-x) + 2f(x) = 3x$$

Find $f(0)$

Question 16

ABCD is a square with sides $7\sqrt{2}$ cm.

E is a point on DC such that $EC = 2\sqrt{2}$ cm and $CF \perp BE$. What is the area of AFED?

QUESTION 17

The mean of n numbers is p . When the number q is removed from the list of numbers averaged, then the mean increases by 2. Find the value of q in terms of n and p .

Question 18

$$2^{96} - 1$$

Has two factors between 60 and 70. What are they?

Question 19

Each male Honey-bee has a single female parent whilst each female honey-bee has both a male and a female parent. In the 10th generation back, only, how many ancestors does a male honey-bee have?

Question 20

What are the next two terms in the following sequence of numbers

1; 1; 2; 1; 3; 1; 2; 4; 1; 5; 1; 2; 3; 6;; .